QP CODE: 25900653

Scheme for Valuation

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third Semester

Discipline Specific Elective (DSE) – MG3DSESTA200

STATISTICAL TECHNIQUES FOR DATA SCIENCE AND MACHINE LEARNING

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Max Marks: 50**

Part A
Short Answer type questions
Answer any 7 questions (Each question carries 2 marks)

1. Any two assumptions of One-way ANOVA (2 Mark)
2. Definition of F distribution (1 Mark)
   Any two applications of F distribution (1 Mark)
3. Any two examples of unsupervised learning (2 Mark)
4. Definition of linear regression (2 Mark)
5. Definition of residuals in regression (2 Mark)
6. Explain any two statistical tools in machine learning. (2 Mark)
7. Definition of time series data (2 Mark)
8. Explain any two examples of data science in transportation (2 Mark)
9. Define biological data (1 Mark)
   Any one feature of biological data (1 Mark)
10. Explain any two impacts of data science (2 Mark)

Part B
Short Essay Questions
Answer any 4 questions (Each question carries 6 marks)

11. Write the null and alternative hypothesis (2 Mark)
    Calculate F statistic (1.193) (2 Mark)
    Find F tabled value (1 Mark)
    Write the conclusion
    (The diets do not differ significantly in their variances.) (1 Mark)
12. Explain in detail the test procedure of t test (6 Mark)
13. Write the null and alternative hypothesis (2 Mark)
    Calculate Chi-square test statistic (1.792) (2 Mark)
    Find tabled value ($\chi^2_{0.05,6}$=12.592) (1 Mark)
    Write the conclusion (Accidents are uniformly distributed over the week) (1 Mark)

14. Write the null and alternative hypothesis (2 Mark)
    Calculate Chi-square test statistic (7.785) (2 Mark)
    Find tabled value ($\chi^2_{0.05,1}$=3.841) (1 Mark)
    Write the conclusion
    (There is a significant association between hair colour and eye colour.) (1 Mark)
15. Differentiate between traditional programming and machine learning (6 Mark)
16. Definition of structured and unstructured data (3 mark)
    Applications of structured and unstructured data (3 mark)

Part C
Essay Questions
Answer any 1 question (Each question carries 12 marks)

17. Definition of logistic regression (2 mark)
    Definition of confusion matrix (2 mark)
    Definition of accuracy (2 mark)
    Definition of precision (2 mark)
    Definition of Recall (2 mark)
    Definition of F1 Score (2 mark)
18. Explain Data Evolution (4 mark)
    Explain in detail the stages of data evolution (2 x 4 =8)
    [Stages: 1. Manual Data Systems
    2.Computerised/Traditional Database systems
    3.Data warehousing and business intelligence systems
    4. Big data Era]